



Phase 1:

Tasks and Responsibilities of a Theorist

Version 0.1

Goal: Achieve consensus on what we think theoretical psychologists (should) do/create

Output: Consensus list of core tasks/responsibilities of the theorists

Introductory note

This is a list of eight core responsibilities of theoretical psychologists. The list identifies tasks that are important for advancing theoretical psychology and psychological science more broadly. These responsibilities do not imply that every theoretical psychologist must undertake all tasks, or that each task must be carried out by one person. Rather, the claim is that these tasks are sufficiently central that at least some theoretical psychologists should take responsibility for them.

The list distinguishes non-delegable theoretical responsibility from delegable execution. Many tasks require collaboration with empirical psychologists, psychometricians, statisticians, computational modelers, software developers, designers, domain specialists, or applied researchers. However, the theoretical responsibility cannot be delegated completely. The theorist remains responsible for clarifying what is being explained, measured, formalized, tested, revised, documented, communicated, or applied, and how this work bears on theory.

Consensus on all eight items was reached by 13 experts participating in the ECTO project, according to the predetermined consensus criterion. Expert feedback was used to refine the wording, clarify role boundaries, identify important but delegable tasks, and improve the distinction between theoretical responsibility and technical, empirical, or infrastructural implementation.



Core tasks of a (modern) theoretical psychologist

1. Identification and formation of explananda

The theorist identifies, categorizes, and orders the phenomena, capacities, or problems to be explained by theory. The theorist distinguishes established explananda from provisional patterns, speculative targets, and raw data patterns, while making their epistemic status explicit.

2. Developing and evaluating construct architecture

The theorist defines, differentiates, coordinates, and, where necessary, prunes constructs; builds ontologies and taxonomies; and specifies how theoretical terms refer to psychological phenomena, capacities, attributes, states, processes, or mechanisms.

3. Coordinating the relation between theory and measurement

The theorist specifies how constructs could, in principle, be operationalized, indexed, measured, or otherwise evidenced; what counts as valid indicators or indices; and what assumptions connect these to theoretical entities. Technical measurement development can be collaborative or delegated.

4. Proposing, specifying, and connecting mechanisms

The theorist proposes and refines mechanistic, functional, or process-based explanations, distinguishing preliminary sketches from more specified accounts and relating psychological explanations across cognitive, affective, motivational, behavioral, bodily, neural, social, developmental, and environmental levels.

5. Constructing theories and their formal representations (models)

The theorist builds verbal, formal, diagrammatic, computational, mathematical, or hybrid theories, models, and model families that specify core propositions, clarify assumptions, support comparison and testing, and remain usable by relevant scientific audiences.

6. Exploring and repairing existing theories.

The theorist extends, constrains, deepens, and repairs theories by resolving tensions, refining mechanisms, examining analogies, developing variants, and situating theories in relation to other theories when this improves coherence, precision, parsimony, explanatory value, scope, or testability.

7. Derivation, explanatory appraisal, scope mapping, and comparison.

The theorist derives implications from theories and models, including explanations of established phenomena, qualitative or quantitative predictions, intervention predictions, counterfactual expectations, and expected failures. The theorist identifies auxiliary assumptions,



maps scope and boundary conditions, compares rival theories, and specifies what evidence would count for or against each theory.

8. Integration, revision, and programme design.

The theorist integrates fragmented empirical, formal, and measurement-related evidence; revises, retains, forks, merges, or discards theories; documents assumptions, versions, and open questions; and helps design theory-guided research programmes. This includes articulating standards for judging theories, including their explanatory value, empirical interface, usability, and potential harms.